4.20.3.6. Sequence Numbers

All PAFTP packets SHALL be sent with sequence numbers, regardless of whether they contain SDU segments (for example, a packet acknowledgement with no attached segment payload). The purpose of sequence numbers is to facilitate the PAFTP receive window. A PAFTP sequence number SHALL be defined as an unsigned 8-bit integer value that monotonically increments by 1 with each packet sent by a given peer. A sequence number incremented past 255 SHALL wrap to zero.

Sequence numbers SHALL be separately defined for either direction of a PAFTP session. The sequence number of the first packet sent by the Commissioner after completion of the PAFTP session handshake SHALL be zero. The Commissionable Device PAFTP handshake response bears an implied sequence number of zero because it occupies a slot in the Commissioner's receive window. The Commissioner acknowledges the Commissionable Device's PAFTP handshake response with an acknowledgement sequence of zero. For this reason, the sequence number of the first data packet sent by the Commissionable Device after completion of the PAFTP session handshake SHALL be 1.

Peers SHALL check to ensure that all received PAFTP packets properly increment the sender's previous sequence number by 1. If this check fails, the peer SHALL close the PAFTP session and report an error to the application.

4.20.3.7. Receive Windows

The purpose of the receive window is to enable flow control at the WFA-USD layer between PAFTP peers.

Flow control is required at the WFA-USD transport layer for embedded platforms that use "minimal" Wi-Fi chipsets supporting WFA-USD. These platforms may have limited space on the host processor to receive packets from their Wi-Fi chipsets. When the Wi-Fi chip sends the result of a received WFA-USD SDF Follow-up message PDU to the host processor, that payload and the corresponding PAFTP packet will be permanently lost if the host does not have enough space to receive it. For this reason, knowledge of a remote host's ability to reliably receive WFA-USD SDF Follow message PDUs is presented at the transport layer in the form of the PAFTP receive window.

Both peers in a PAFTP session SHALL define a receive window, where the window's size indicates the number of WFA-USD SDF Follow-up message PDUs (that is, PAFTP segments) a peer can reliably receive and store without session-layer acknowledgment. A maximum window size SHALL be established for both peers as part of the PAFTP session handshake. To prevent sequence number wrap-around, the largest maximum window size any peer may support is 255.

Both peers SHALL maintain a counter to reflect the current size of the remote peer's receive window. Each peer SHALL decrement this counter when it sends a packet and increment this counter when a sent packet is acknowledged.

If a local peer's counter for a remote peer's receive window is zero, the window SHALL be considered closed, and the local peer SHALL NOT send packets until the window reopens (is incremented above zero). When a closed window reopens, a local peer SHALL immediately resume any pending PAFTP packet transmission.

A local peer SHALL also not send packets if the remote peer's receive window has one slot open and the local peer does not have a pending packet acknowledgement. This is to avoid the situation